

SIMATS ENGINEERING — ASSESSMENT TOOL 3

COURSE CODE:	MLA03	COURSE NAME:	Reinforcement learning for Environment and Agent
COURSE OUTCOME:	CO5: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)		
FACULTY:	Dr. G. A. SENTHIL	ASSESSMENT WEIGHTAGE / MARKS:	Weightage: 30% Total Marks: 50 Duration: 1 hr

Annexure A — Analytical Problem-Solving Assessment

SECTION 1: MARKOV DECISION PROCESSES & DYNAMIC PROGRAMMING

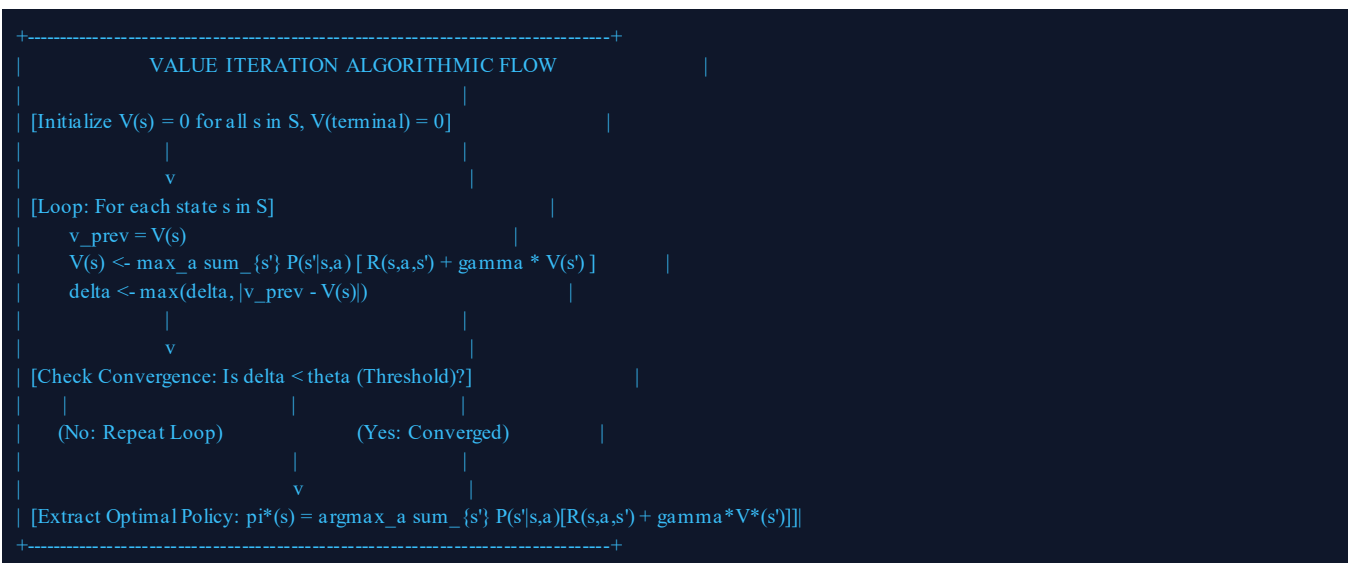
Question 1: Warehouse Robot Optimal Path Planning using Dynamic Programming & Value Iteration (10 Marks)

Problem Statement: A warehouse robot must determine the shortest path to transport goods while avoiding static/dynamic obstacles. Apply Dynamic Programming to develop an optimal policy and explain how Value Iteration improves navigation efficiency.

Solution & Theoretical Formulation:

1. MDP Modeling for Warehouse Grid:

- **State Space (S):** Grid coordinates (x, y) representing robot locations, obstacles, and docking stations.
- **Action Space (A):** Discrete movements {North, South, East, West, Stay}.
- **Transition Model $P(s'|s, a)$:** Probability of reaching target cell s' given action a (deterministic in structured warehouses or stochastic with slip factor).
- **Reward Function $R(s, a, s')$:** +100 on reaching destination, -50 on obstacle collision, and -1 step cost per movement to encourage shortest path.



2. Value Iteration Efficiency & Navigation Impact:

- **Bellman Optimality Backup:** Value iteration directly combines policy evaluation and policy improvement into a single step without waiting for exact policy evaluation convergence in each iteration.
- **Global Path Optimality:** By updating the value function using the max operator over all legal actions, the robot avoids local minima and computes the shortest, obstacle-free trajectory to target drop-off zones.

SECTION 2: TEMPORAL DIFFERENCE CONTROL IN UNCERTAIN ENVIRONMENTS

Question 2: SARSA-Based Action-Value Learning for Autonomous Delivery Robots (10 Marks)

Problem Statement: A delivery robot operates in an uncertain environment with dynamic obstacles and slippery paths. Apply the SARSA algorithm to update the action-value function and explain how the learned policy adapts to environmental changes.

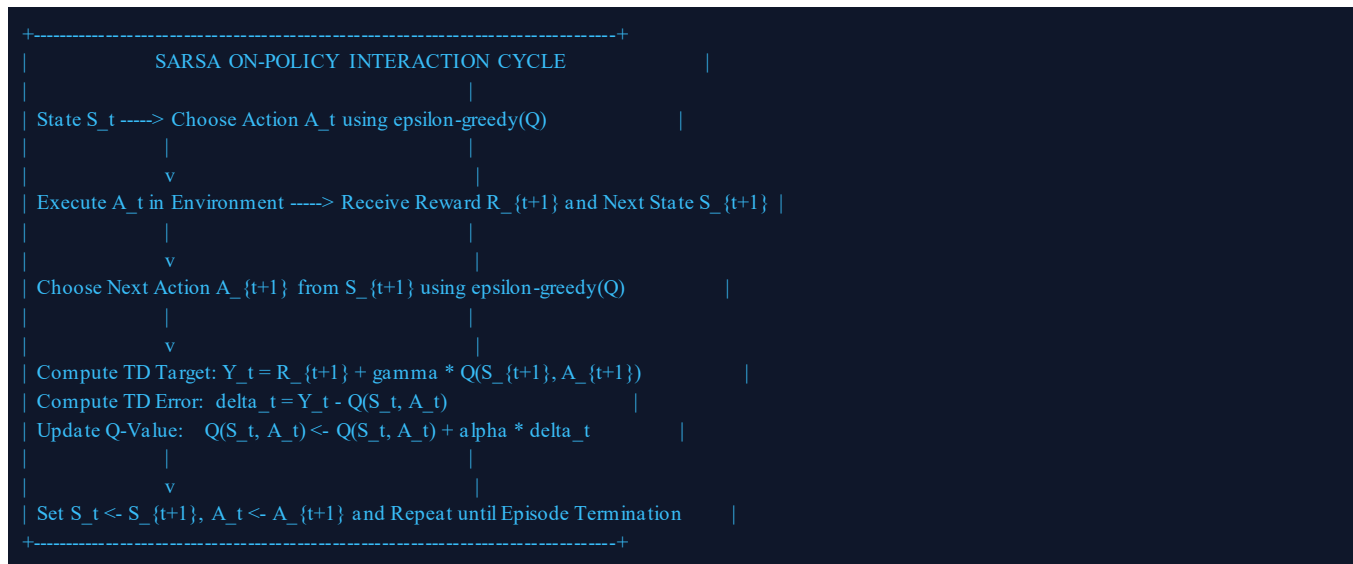
Solution & Theoretical Formulation:

1. SARSA On-Policy Learning Update Rule:

SARSA (State-Action-Reward-State-Action) is an on-policy TD control algorithm that updates the action-value function $Q(s, a)$ using the actual transition taken by the behavior policy:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)]$$

where α is the learning rate, γ is the discount factor, and A_{t+1} is selected using an ϵ -greedy policy: $A_{t+1} \sim \pi(\cdot | S_{t+1})$.



2. Environmental Adaptation & Safety Analysis:

- **On-Policy Safety Advantage:** Unlike Q-learning (which assumes greedy actions in the backup), SARSA accounts for the exploratory actions actually taken. In dangerous delivery routes (e.g., cliff paths or slippery terrain), SARSA learns safer paths that maintain clearance from hazards.
- **Dynamic Adaptation:** When physical terrain changes or obstacles move, negative rewards update $Q(s, a)$ locally, naturally shifting the ϵ -greedy policy to alternate viable detours.

SECTION 3: POLICY GRADIENT METHODS FOR CONTINUOUS CONTROL

Question 3: REINFORCE Algorithm for Continuous Robotic Arm Object Manipulation (10 Marks)

Problem Statement: A multi-joint robotic arm performs continuous object grasping and manipulation. Apply the REINFORCE (Monte Carlo Policy Gradient) algorithm to develop a parameterized policy that maximizes task completion accuracy.

Solution & Theoretical Formulation:

1. Parameterized Continuous Policy & Likelihood Ratio:

Continuous joint torques $a \in \mathbb{R}^d$ are generated by parameterizing the policy as a Gaussian distribution: $a \sim \pi_\theta(a|s) = \mathcal{N}(\mu_\theta(s), \Sigma_\theta(s))$.

The policy gradient theorem defines objective optimization $J(\theta) = E_{\tau \sim \pi_\theta} [G_0]$ as:

$$\nabla_\theta J(\theta) = E_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t | s_t) \cdot G_t \right]$$

where return $G_t = \sum_{k=t}^{T-1} \gamma^k R_{k+1}$.

```
+-----+
|          REINFORCE WITH BASELINE FOR CONTINUOUS ROBOTIC CONTROL          |
+-----+
| 1. Collect Episode Rollout:  $\tau = (s_0, a_0, r_1, s_1, a_1, \dots, s_T) \sim \pi_\theta$  |
|                                     |
| 2. For each time step  $t = 0, 1, \dots, T-1$ : |
|   • Compute Discounted Return:  $G_t = \sum_{k=t}^{T-1} \gamma^k r_{k+1}$  |
|   • Compute State Baseline Value:  $V_\phi(s_t)$  |
|   • Compute Advantage / Scaled Return:  $\delta_t = G_t - V_\phi(s_t)$  |
|                                     |
| 3. Accumulate Policy Gradient: |
|    $\text{grad}_\theta J \leftarrow \text{grad}_\theta \log \pi_\theta(a_t | s_t) * (G_t - V_\phi(s_t))$  |
|                                     |
| 4. Update Policy & Baseline Parameters: |
|    $\theta \leftarrow \theta + \alpha_\theta * \text{grad}_\theta J$  |
|    $\phi \leftarrow \phi - \alpha_\phi * \text{grad}_\phi \| V_\phi(s_t) - G_t \|^2$  |
+-----+
```

2. Variance Reduction via Baseline:

Robotic manipulation has high degrees of freedom, leading to extreme variance in Monte Carlo returns. Incorporating a state-value baseline $V_\phi(s_t)$ leaves the expected gradient unbiased while drastically reducing variance: $E[\nabla_\theta \log \pi_\theta(a_t | s_t) V(s_t)] = 0$.

SECTION 4: ADVANCED DEEP Q-NETWORKS ARCHITECTURES

Question 4: Comparative Analysis: Standard DQN, Double DQN (DDQN), and Dueling DQN (10 Marks)

Problem Statement: A complex game-playing AI and autonomous vehicle driving controller require value-based deep RL. Analyze the structural innovations and performance advantages of Double DQN (DDQN) and Dueling DQN over the standard DQN baseline.

Feature / Architecture	Standard DQN (Mnih et al.)	Double DQN (DDQN)	Dueling DQN (Wang et al.)
Target Value Formulation	$Y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$	$Y = r + \gamma Q(s', \arg\max_{a'} Q(s', a'; \theta); \theta^-)$	$Y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$ with dual streams

Core Architectural Modification	Single deep neural network mapping state s to action values $Q(s, a)$.	Decoupled action selection (online θ) from action evaluation (target θ^-).	Network bifurcates into State Value $V(s; \theta, \beta)$ and Advantage $A(s, a; \theta, \alpha)$.
Overestimation Bias Mitigation	Suffers from severe maximization bias (overestimates Q -values).	Completely eliminates overestimation bias, improving policy stability.	Reduces variance in states where actions have identical long-term impact.
Convergence & Performance	Slow and prone to oscillatory divergence in high dimensions.	Faster convergence with realistic and stable value bounds.	Superior policy evaluation sample efficiency in complex state spaces.

Dueling Network Aggregation Identity: $Q(s, a; \theta, \alpha, \beta) = V(s; \theta, \beta) + [A(s, a; \theta, \alpha) - (1 / |A|) \sum_{a'} \{a'\} A(s, a'; \theta, \alpha)]$. Subtracting the mean advantage ensures unique identifiability of $V(s)$ and $A(s, a)$.

SECTION 5: ADVANCED ACTOR-CRITIC (PPO, TRPO, DDPG) & MULTI-AGENT RL

Question 5: Continuous Robotic Control (PPO/TRPO/DDPG) & Multi-Agent Traffic Coordination (10 Marks)

Problem Statement: Evaluate the suitability of PPO, TRPO, and DDPG for continuous robotic control, and analyze how Multi-Agent RL (MARL) coordinates smart city traffic management across scalability, non-stationarity, and fairness.

Algorithm	Key Mechanism / Constraint	Sample Efficiency & Stability	Computational Complexity
TRPO (Trust Region Policy Opt.)	Enforces hard KL-divergence constraint: $E[D_{KL}(\pi_{old} \pi_{\theta})] \leq \delta$ via Conjugate Gradient.	Highly stable; monotonic improvement guarantee. Moderate sample efficiency.	Very High (requires Hessian-vector products and Fisher Information Matrix).
PPO (Proximal Policy Opt.)	Clipped surrogate objective: $L^{CLIP}(\theta) = E[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)]$.	High stability; superior empirical sample efficiency.	Low / First-order (uses standard SGD/Adam; easy to implement and scale).
DDPG (Deep Deterministic PG)	Off-policy Actor-Critic for continuous action deterministic policy $a = \mu_{\theta}(s)$ with replay buffer.	Sample efficient due to replay buffer, but sensitive to hyperparameter tuning.	Moderate (susceptible to Q-overestimation; addressed by TD3).

Multi-Agent RL (MARL) for Smart Traffic Management Analysis:

- **Centralized Training with Decentralized Execution (CTDE):** Traffic light agents communicate during training via a centralized critic observing global traffic states, while executing decentralized policies at individual intersections using local sensor inputs.
- **Non-Stationarity Mitigation:** Because agents update policies simultaneously, the environment appears non-stationary. CTDE and value-factorization methods (e.g., QMIX, MAPPO) stabilize joint convergence.
- **Fairness & Ethical Objectives:** Reward functions balance total vehicle throughput with maximum wait-time thresholds to prevent traffic starvation on minor arterial roads.



Code of Conduct:

I **K. Balamurali / 192472010** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

NAME: K. Balamurali

REG NO: 192472010

SIGNATURE OF THE STUDENT: K. Balamurali

DATE: 25-08-2026

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively